Unit 3 – Cloud Security Assignment

Topic: Set Up a Web Application on Amazon EC2

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1. Aim

To launch an Amazon EC2 instance, secure it with Security Groups, connect to it over SSH, install a web server, deploy a simple HTML web application and verify it in a browser.

2. Requirements

- An active AWS account (Free Tier is enough).
- A key pair (.pem file) for SSH login.
- A terminal or SSH client (Git Bash was used on Windows).
- A web browser to test the website.

3. Procedure

Step 1: Launch an EC2 instance on AWS

- Log in to the AWS Management Console, select the region (Asia Pacific – Mumbai) and open the EC2 service.
- Click "Launch instance" and enter the name "web-server".
- Select the AMI: Amazon Linux 2023 (Free tier eligible).
- Select the instance type: t2.micro (1 vCPU, 1 GiB memory).
- Select the key pair "my-key-pair" (created earlier in RSA .pem format).
- Keep the default VPC and subnet and set "Auto-assign public IP" to Enable.
- Choose "Create security group" (configured in Step 2).

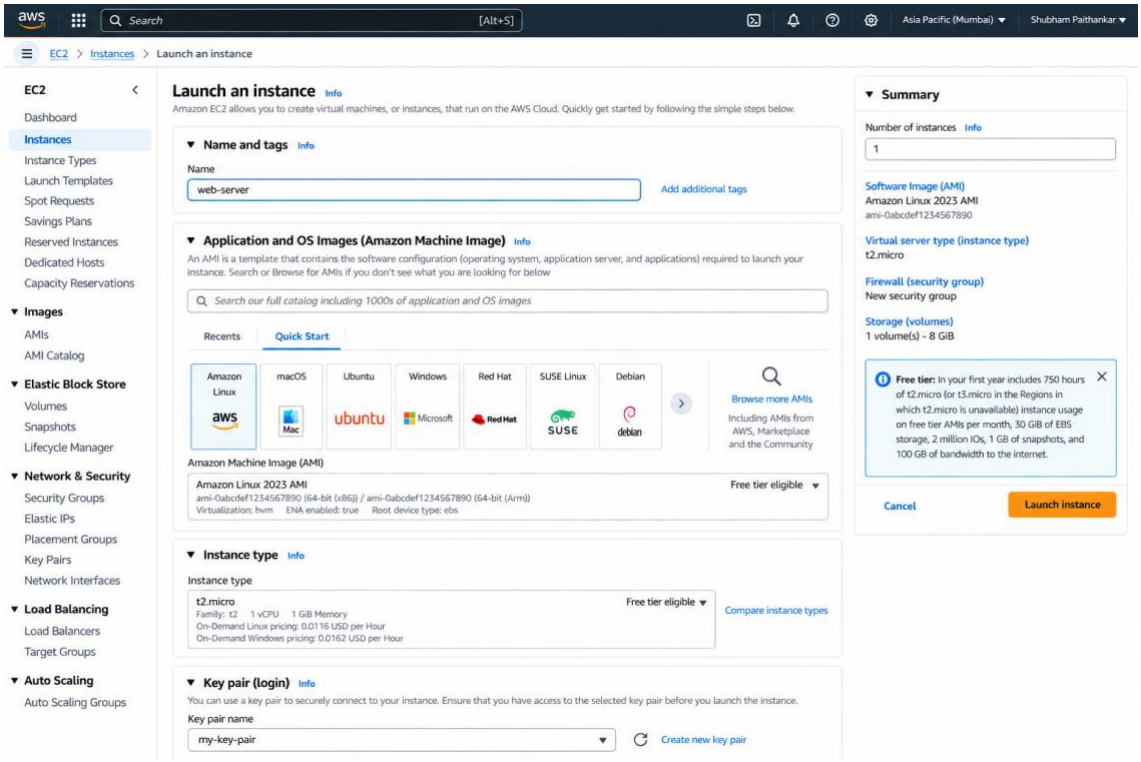


Figure 1: Launch instance – name, AMI, instance type and key pair

Observation: The instance is named web-server and uses the Amazon Linux 2023 AMI (free tier eligible) with the t2.micro instance type. The key pair my-key-pair is selected for login. The summary panel on the right shows 1 instance, 8 GiB storage and a new security group.

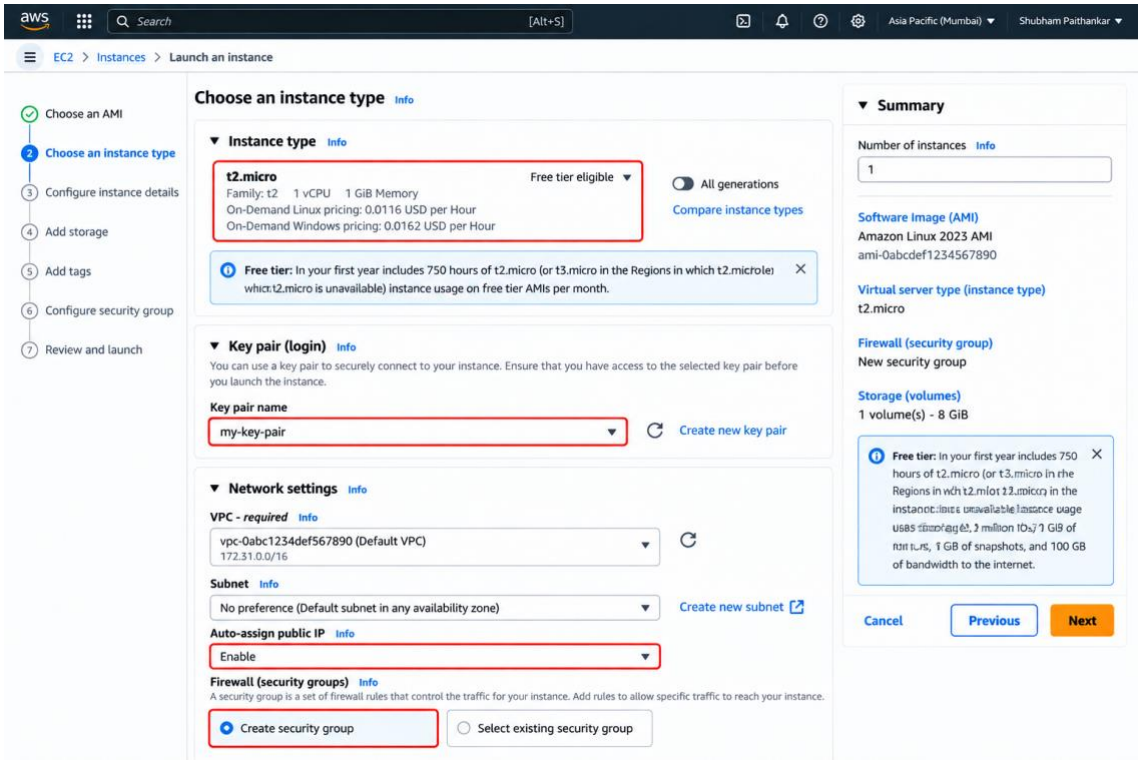


Figure 2: Instance type, key pair and network settings

Observation: t2.micro and my-key-pair are selected, the default VPC (172.31.0.0/16) is used, Auto-assign public IP is set to Enable so that the server gets a public address, and the option "Create security group" is selected.

Step 2: Configure Security Groups

A Security Group acts as a virtual firewall for the instance. A new security group named web-sg was created with the following inbound rules:

Type	Protocol	Port	Source	Purpose
SSH	TCP	22	My IP (single IP, /32)	Remote login to the server
HTTP	TCP	80	0.0.0.0/0 (Anywhere)	Public access to the website

Port 22 is restricted to a single IP address (my own) so that nobody else on the internet can try to log in. Port 80 is open to everyone so that visitors can view the web application. No other ports are opened (principle of least privilege).

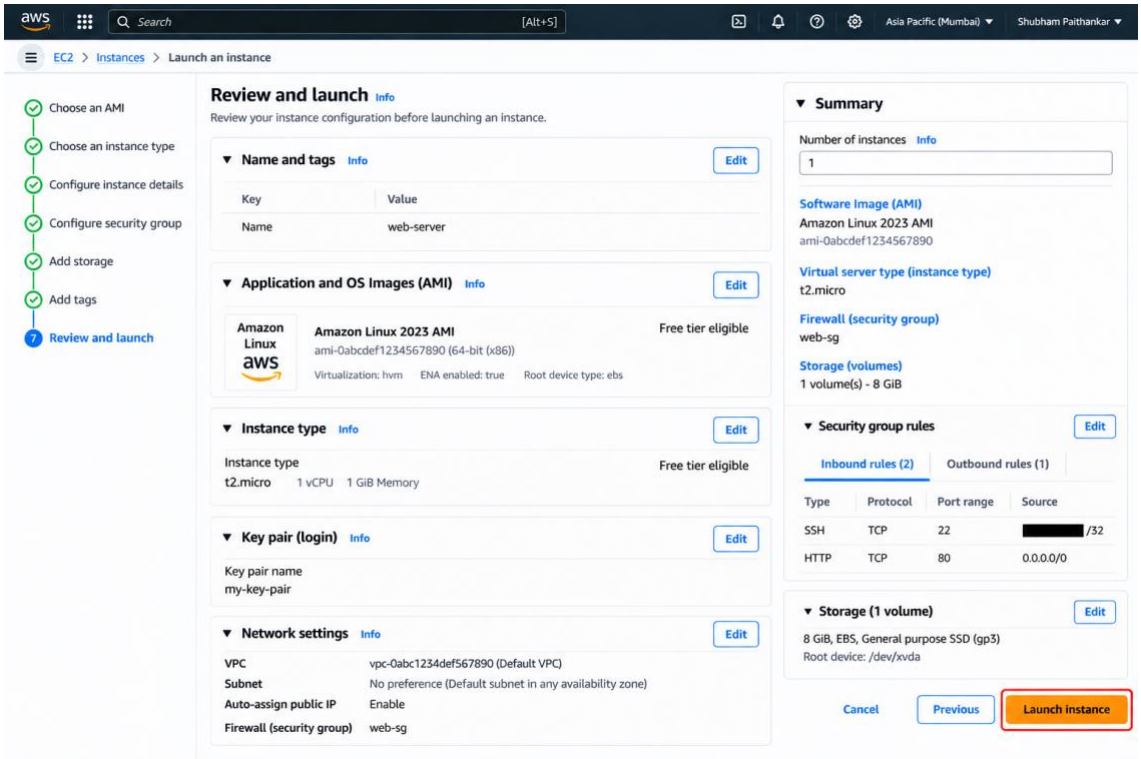


Figure 3: Review and launch – security group web-sg with inbound rules

Observation: The review page shows the security group web-sg with 2 inbound rules: SSH (TCP 22) from a single IP address ending in /32 (the IP is hidden in the screenshot for privacy) and HTTP (TCP 80) from 0.0.0.0/0. The storage is 8 GiB gp3. After checking everything, "Launch instance" was clicked.

After launching, the instance was checked in the EC2 Instances list:

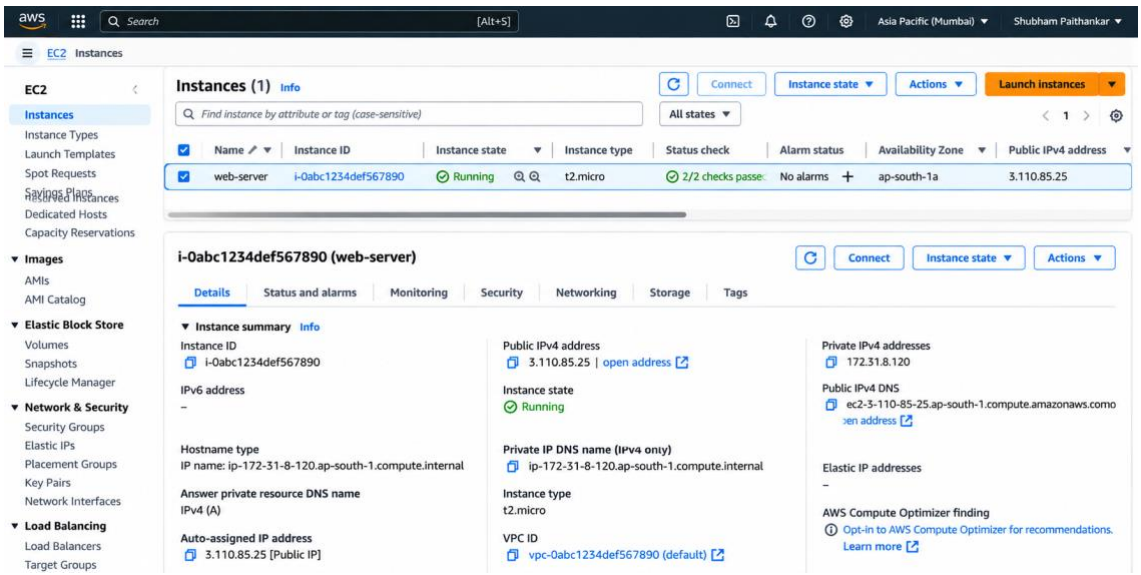


Figure 4: Instance in Running state

Observation: The instance web-server (t2.micro, Availability Zone ap-south-1a) shows Instance state "Running" and "2/2 checks passed". It has the public IPv4 address 3.110.85.25

and private IPv4 address 172.31.8.120. The public IP is used for SSH and for opening the website.

Step 3: Connect to the instance using SSH

The terminal (Git Bash) was opened in the folder where the key file is saved and the following command was run:

```
ssh -i "web-key.pem" ec2-user@3.110.85.25
```

On the first connection the message "authenticity of host can't be established" appears. Typing "yes" adds the server to the list of known hosts. The prompt then changes to ec2-user@ip-172-31-8-120, which shows the connection is successful.

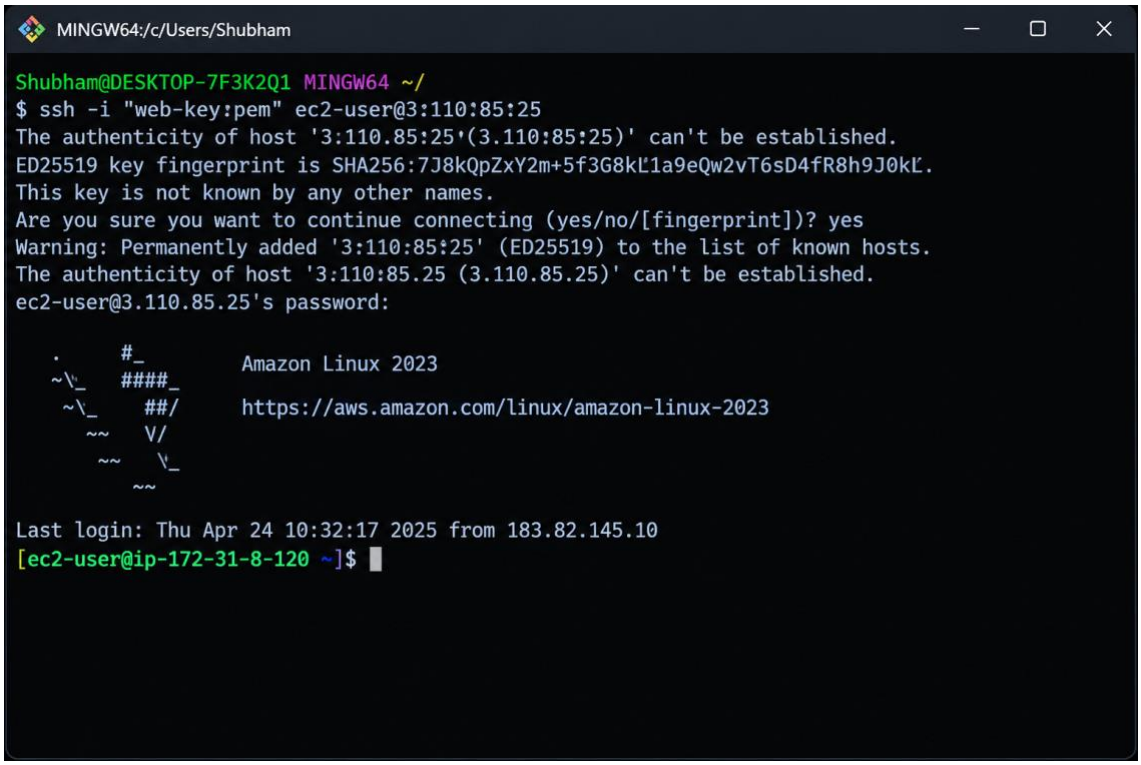


Figure 5: Successful SSH connection to the EC2 instance

Observation: The Amazon Linux 2023 welcome banner is displayed and the prompt [ec2-user@ip-172-31-8-120 ~]$ appears, so the login to the instance is successful.

Step 4: Install and configure Apache web server

```
sudo dnf update -y
sudo dnf install httpd -y
sudo systemctl enable --now httpd
sudo systemctl status httpd
```

The "enable --now" option starts Apache immediately and also makes it start automatically whenever the instance reboots.

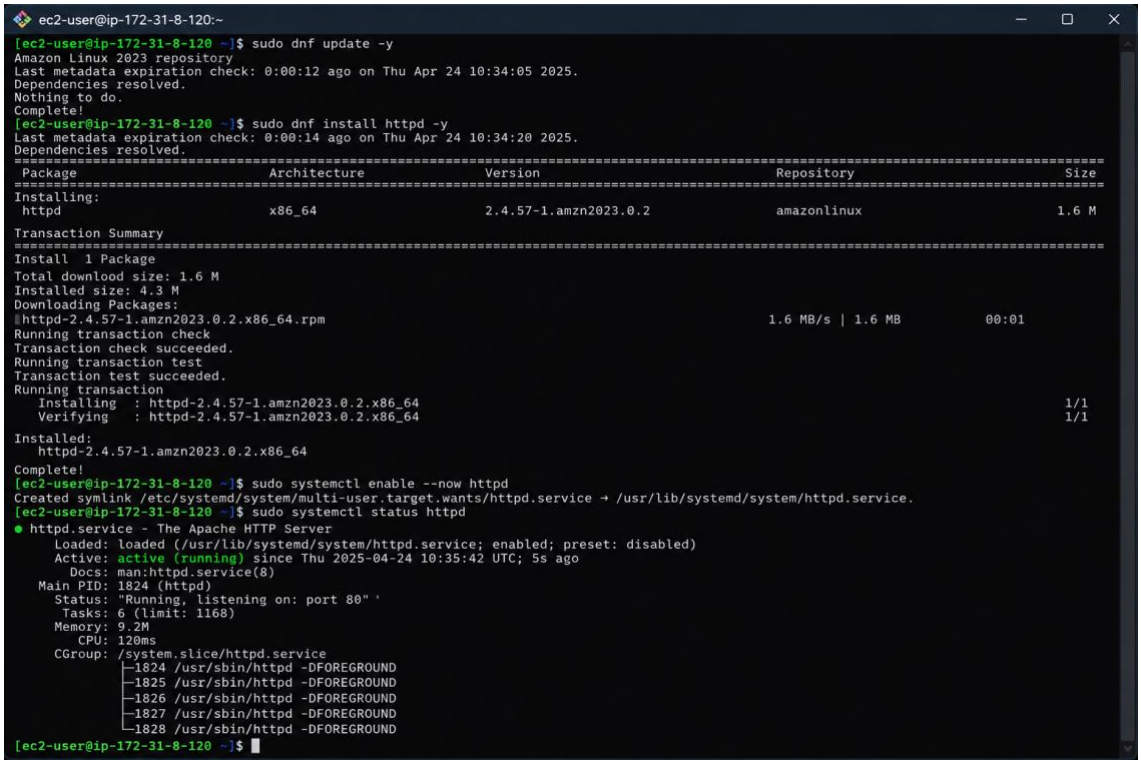


Figure 6: Apache (httpd) installed and running

Observation: The system was already up to date ("Nothing to do"). Package httpd-2.4.57 was installed from the amazonlinux repository. The status command shows "active (running)" with the message "Running, listening on port 80".

Step 5: Deploy a simple HTML web application and verify it

The default web root of Apache is /var/www/html. The index.html file was created there using the tee command:

```
sudo tee /var/www/html/index.html <<'EOF'
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>AWS EC2 Web Application</title>
  <style>
    body { background-color: #101820; color: #00ff99;
           font-family: Arial, sans-serif; text-align: center;
           padding-top: 80px; }
    h1 { font-size: 40px; }
    p  { font-size: 20px; }
  </style>
</head>
<body>
  <h1>Welcome to My AWS EC2 Web Application</h1>
  <p>Successfully deployed using Apache on Amazon EC2.</p>
</body>
</html>
EOF
```

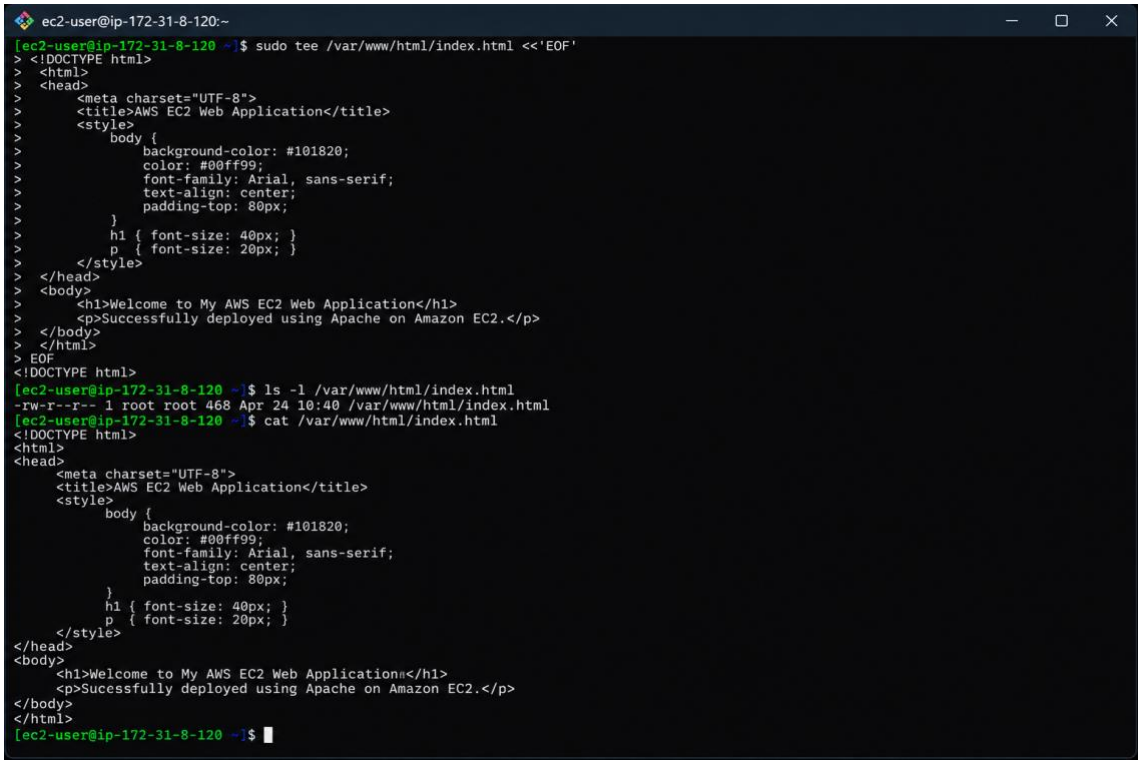


Figure 7: Creating and checking index.html

Observation: The file /var/www/html/index.html was created with sudo tee. The ls -l command shows the file (468 bytes, permissions -rw-r--r--) and cat displays its HTML content.

Finally, a browser was opened and the public IP address of the instance (http://3.110.85.25) was visited.

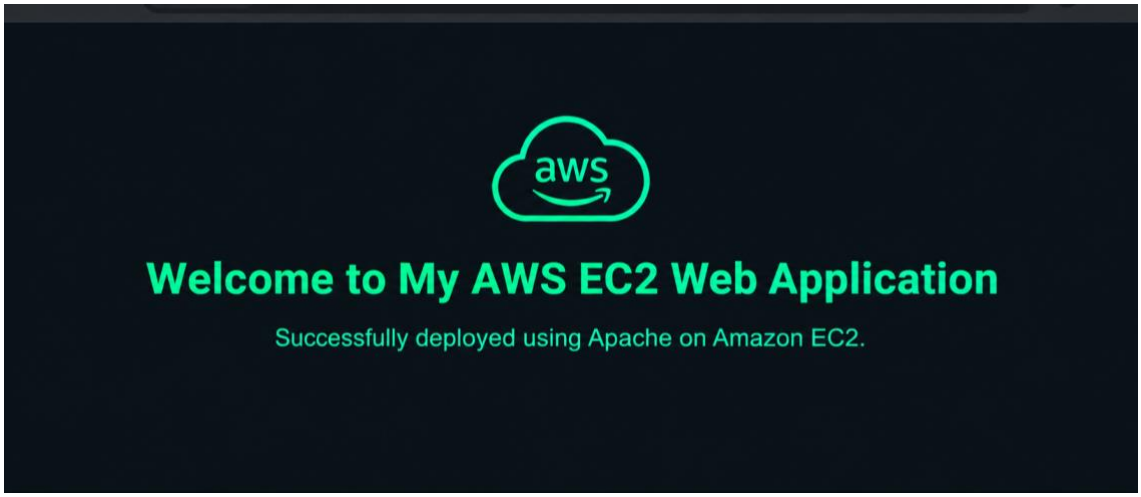


Figure 8: Web application opened in the browser using the public IP

Observation: The page titled "AWS EC2 Web Application" loads from 3.110.85.25 and shows "Welcome to My AWS EC2 Web Application – Successfully deployed using Apache on Amazon EC2." The browser shows "Not secure" only because the site uses HTTP (port 80) and no SSL certificate is installed.

4. Security Configuration Summary

- SSH (port 22) allowed only from a single IP address (/32), not from 0.0.0.0/0.
- HTTP (port 80) allowed publicly, as required for a website.
- Login is done with a private key pair (.pem file).
- The .pem key file is kept private and is never shared.
- System packages were updated with dnf update before installing software.
- Only the required service (Apache) is installed and no other ports are open.
- For a real production site, HTTPS (port 443) with an SSL certificate should be added.

5. Result

An EC2 instance was launched, secured with a Security Group, accessed through SSH, and configured with the Apache web server. A simple HTML web application was deployed and successfully verified in the browser using the instance's public IP address.

6. Conclusion

This assignment showed how a web application can be hosted on the cloud using Amazon EC2. It also showed the importance of Security Groups: allowing SSH only from a trusted IP and exposing only the required web port reduces the chances of unauthorized access.